

Connecting to the microROS Agent

1. Objectives of This Section

Understand the role of the micro-ROS Agent, and learn how to start the Agent and confirm the communication between the control board and the virtual machine.

2. Explanation of Key Terms

Keyword	Description
micro-ROS Agent	A bridging program running on the virtual machine that communicates ROS2 topics with the micro-ROS client on the ESP32-S3 control board via UDP
UDP port	The network port that the Agent listens on; the default is <code>8090</code> , and it must be consistent with the control board configuration
Green light	The green LED on the control board; a steady light means the Agent has been successfully connected

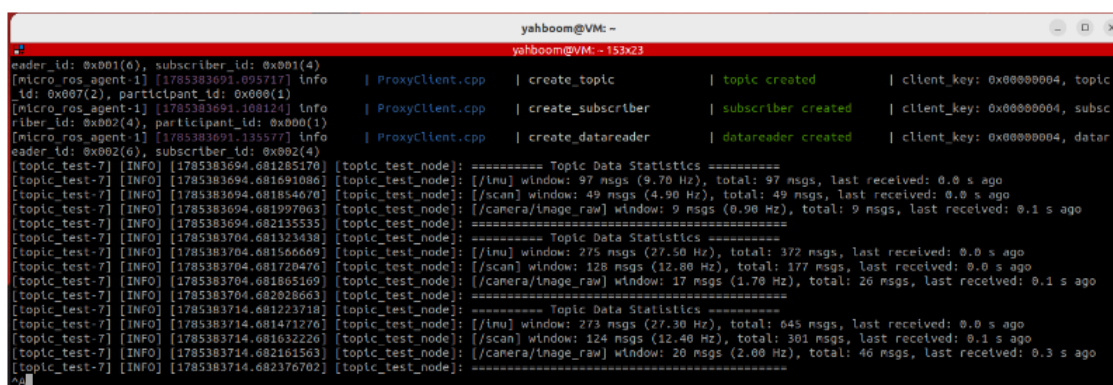
3. Prerequisites

1. The robot parameter configuration (Wi-Fi, Agent IP, port, etc.) has been completed. Refer to **4. Virtual Machine Configuration and Robot Connection**.
2. The control board and the virtual machine are on the same LAN.

4. Starting the micro-ROS Agent

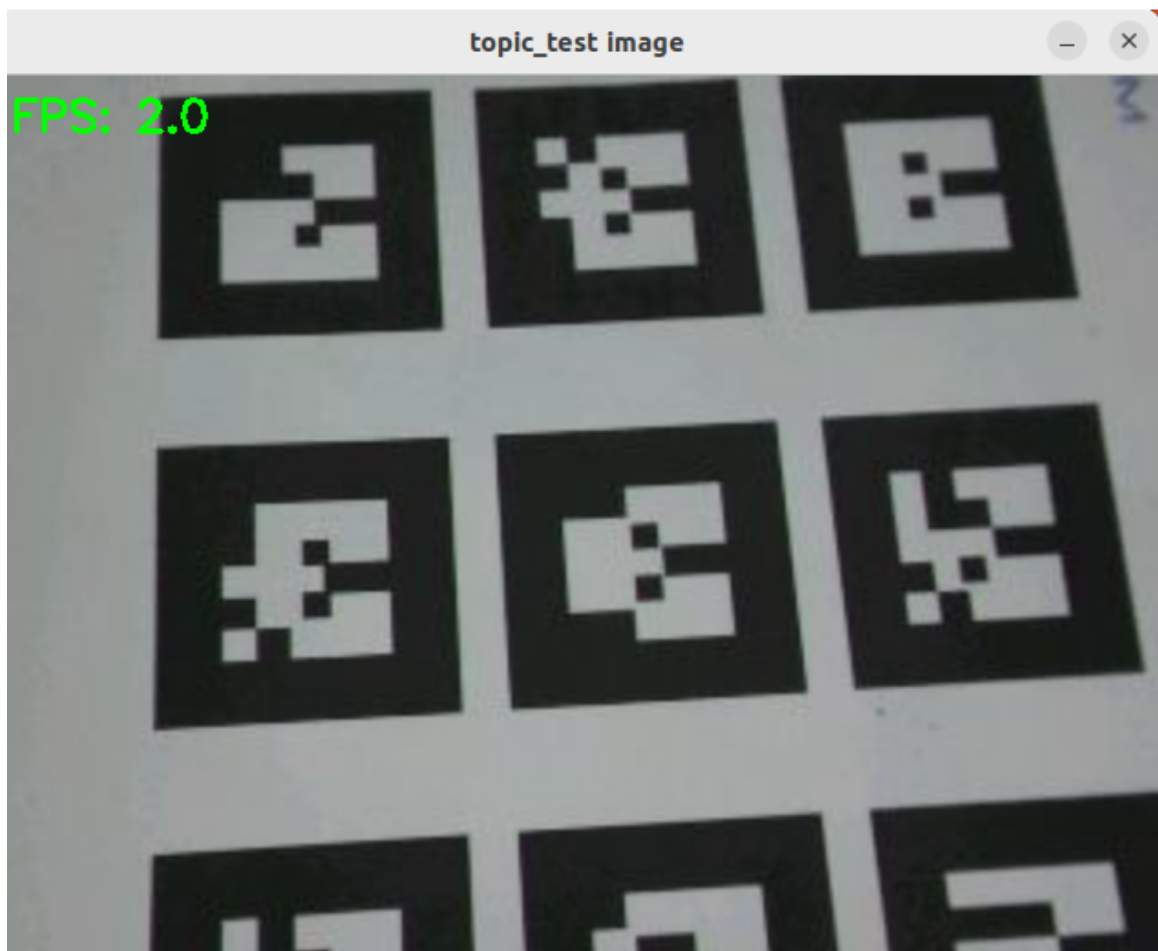
Run the following in `MicroROS_V2_VM`:

```
ros2 launch microros_v2_bringup test_connect.launch.py
```



```
yahboom@VM: -
yahboom@VM: ~ 153x23
[header_id: 0x001(6), subscriber_id: 0x001(4)]
[micro_ros_agent-1] [1785383691.095717] info | ProxyClient.cpp | create_topic | topic created | client_key: 0x00000004, topic
_id: 0x007(2), participant_id: 0x000(1)
[micro_ros_agent-1] [1785383691.108124] info | ProxyClient.cpp | create_subscriber | subscriber created | client_key: 0x00000004, subsc
riber_id: 0x002(4), participant_id: 0x000(1)
[micro_ros_agent-1] [1785383691.135577] info | ProxyClient.cpp | create_datareader | datareader created | client_key: 0x00000004, datar
eader_id: 0x002(6), subscriber_id: 0x002(4)
[topic_test-7] [INFO] [1785383694.681285170] [topic_test_node]: ===== Topic Data Statistics =====
[topic_test-7] [INFO] [1785383694.681691086] [topic_test_node]: [/lnu] window: 97 msgs (9.70 Hz), total: 97 msgs, last received: 0.0 s ago
[topic_test-7] [INFO] [1785383694.681854670] [topic_test_node]: [/scan] window: 49 msgs (4.90 Hz), total: 49 msgs, last received: 0.0 s ago
[topic_test-7] [INFO] [1785383694.681907063] [topic_test_node]: [/camera/image_raw] window: 9 msgs (0.90 Hz), total: 9 msgs, last received: 0.1 s ago
[topic_test-7] [INFO] [1785383694.682135535] [topic_test_node]: ===== Topic Data Statistics =====
[topic_test-7] [INFO] [1785383704.681323438] [topic_test_node]: ===== Topic Data Statistics =====
[topic_test-7] [INFO] [1785383704.681566669] [topic_test_node]: [/lnu] window: 275 msgs (27.50 Hz), total: 372 msgs, last received: 0.0 s ago
[topic_test-7] [INFO] [1785383704.681720470] [topic_test_node]: [/scan] window: 128 msgs (12.80 Hz), total: 177 msgs, last received: 0.0 s ago
[topic_test-7] [INFO] [1785383704.681865169] [topic_test_node]: [/camera/image_raw] window: 17 msgs (1.70 Hz), total: 26 msgs, last received: 0.1 s ago
[topic_test-7] [INFO] [1785383704.682028663] [topic_test_node]: ===== Topic Data Statistics =====
[topic_test-7] [INFO] [1785383714.681223718] [topic_test_node]: ===== Topic Data Statistics =====
[topic_test-7] [INFO] [1785383714.681471270] [topic_test_node]: [/lnu] window: 273 msgs (27.30 Hz), total: 645 msgs, last received: 0.0 s ago
[topic_test-7] [INFO] [1785383714.681632226] [topic_test_node]: [/scan] window: 124 msgs (12.40 Hz), total: 301 msgs, last received: 0.1 s ago
[topic_test-7] [INFO] [1785383714.682161563] [topic_test_node]: [/camera/image_raw] window: 20 msgs (2.00 Hz), total: 46 msgs, last received: 0.3 s ago
[topic_test-7] [INFO] [1785383714.682376702] [topic_test_node]: ===== Topic Data Statistics =====
```

After starting the Agent, observe the **green light** on the control board, and you should be able to see the camera feed:



Green light status	Meaning
Steady on	Successfully connected to the Agent; communication is normal
Off	Not connected to the Agent

If the green light is off, check the following:

1. Whether the Wi-Fi name and password of the control board are correct
2. Whether the Agent IP is set to the real IP of the virtual machine
3. Whether the Agent port is consistent (default 8090)
4. Whether the ROS_DOMAIN_ID matches the one set in the virtual machine
5. Whether the virtual machine firewall allows UDP port 8090